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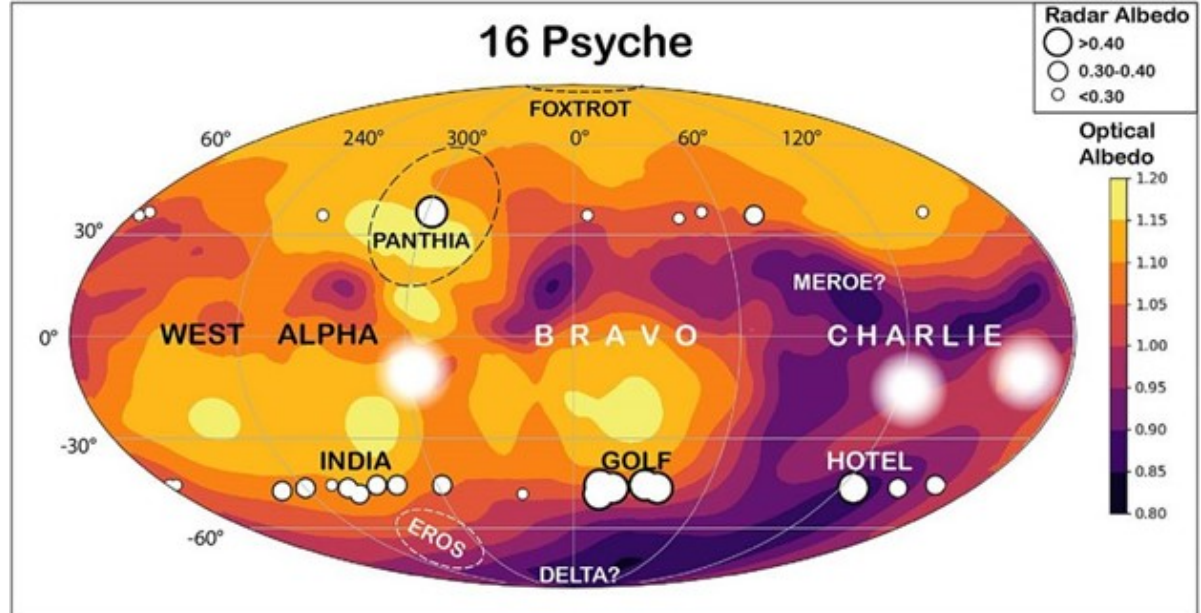
## Cartels and Collapse – The Deep (Economic) Impact of Asteroid Mining

Written By: Finn O'Connor

The IPCC (Intergovernmental Panel on Climate Change) report released in late March has once again spurred discussion on how we can reconcile humanity's massive consumption with the relative continuation of our way of life (1). One solution that has long been

discussed is the possibility of harnessing resources outside the terrestrial ecosystem to bolster our scarce supply. The most attractive place to start this process is with asteroids; mineral-rich rocks that orbit our sun. Companies like Planetary Resources and Deep Space Industries cropped up in recent years intending to harvest asteroids for their resources and transport them back to Earth. However, they have barely begun the exploratory stage, where probes are sent to assess the composition of target asteroids (2).

Exploration is the goal of NASA's latest mission, optimistically slated to launch by the end of 2023, to explore deposits located on *Psyche*, a 226-kilometer-wide hunk of space rock located between Mars and Jupiter (3). Rough estimates already place the value of the nickel alone at nearly \$10,000 quadrillion in (and this is very important) today's value. The mission generated renewed interest in the prospects of asteroid mining, as the implications of being able to extract any resource in such quantity are hard to understate. Subsequent analysis of off-world resource-gathering identifies potential consequences that range from the apocalyptic to the utopian. The purpose of this paper is to parse through how a hypothetical market for asteroid resources could form, and what it might eventually look like based on the contingencies that must be met to access space-borne minerals. We then pivot to look at the effect that bringing such an abundance of resources could have on the global economy. In summary, this paper hopes to identify some of the key challenges that might arise from a potential asteroid-mining industry.



*A map of Psyche that combines optical and radar albedo measurements. The brightest spots might be evidence of rare ferro volcanic activity. (4)*

### **"Asteroid Cartels" and Government Regulation**

To start, the minerals on asteroids such as Psyche are presently considered contingent resources. That is an estimated quantity of a resource that is not yet recoverable due to one or more contingencies. The contingencies, in this case, are, put broadly, the ability to reliably extract resources from a rock floating in space, and then to transport said resources back to Earth at a cheap enough cost to justify the entire project.

To justify a mining project, to make it financially viable, a firm must overcome the exorbitant costs of intra-stellar transportation. NASA's mission to Psyche, which is only exploratory, is projected to cost around \$283.7 million to launch and operate, not including the nearly \$700 million for developing an exploratory spacecraft (5). To develop, build, and launch a craft would cost well above that mark.

The high barriers to entry for the off-world resources market, carry a risk that a monopoly, or even a cartel, could form. A recent in-depth analysis of the effects of asteroid mining by Zhang et al. determined

that only a select few countries with leading aerospace technology could access the asteroid resource market, whereas the remaining countries act only as consumers of celestial resources (6). While we lack the necessary information on any technology that could be utilized in asteroid mining to accurately predict the relationship between fixed and variable costs, we can look at the significant difference between the cost of building Psyche's exploration craft, and its relatively lower operating costs. Therefore, based on what we know now about aerospace technology, it is reasonable to project that the market's fixed costs would be high enough compared to its relative costs to cause a natural monopoly to arise. On top of this, as we will explore later, it is in firms' best interests to control supply to maximize profits. In this case, price fixing might even be necessary. Thus, it is possible that an asteroid-mining cartel forms.

This risk is compounded by the fact that states might have very limited if any, regulatory influence in space. Zhang et al.'s conclusion on global equity is optimistic considering it is based on the assumption that national space agencies like NASA and the CNSA will continue to dominate space exploration and eventual expansion, giving them a high degree of influence over an asteroid-mining industry. Of course, this is not entirely impossible, some of the biggest extractive industries (EIs) consist of the either firm that is majority owned by governments or agencies that operate as an arm of a state. However, EIs on Earth are constrained by the fact that every square inch of the planet outside of international waters is owned by one country or another. A "host country" controls who gets to access its resources, meaning that a government will always have fundamental ownership over EIs. Additionally, while there exist some international agencies that set rules for the activity in low Earth orbit (LEO) and beyond, these are almost entirely normative with little built-in enforcement mechanisms, not to mention they are fairly outdated (the Outer Space Treaty that covers commercial activity in space was signed in 1967) and frustratingly vague. On the other hand, there are no functional means through which a country could claim an asteroid and its resources.

Furthermore, we have seen a recent rise in private space enterprises, such as the ones mentioned at the beginning of this paper, that operate autonomously from national agencies. NASA remains dominant in long-range space exploration, but this is because exploratory missions still lack any profit incentive, requiring government funding to facilitate. But once more tangible benefits to long-range missions materialize, private actors might find a reason to fund their own. This is evident in the rise of “space tourism” which motivates commercial firms to move into NASA’s LEO monopoly (7). The 2015 SPACE (Spurring Private Aerospace and Competitive and Entrepreneurship) Act also affirms that US companies have the right to exploit celestial resources, while also renouncing any jurisdiction over those resources.

All this is to say that we cannot rely on states to have any significant presence in an asteroid-resource market, so traditional anti-trust methods might be ineffective. For example, in Ontario, while the land is controlled by the property owner, the provincial government owns the minerals underneath. Therefore, a mining company must apply for a time-limited lease to develop a deposit, even then the amount extracted is subject to government approval (8). Since there is no existing mechanism for a state to claim an asteroid, there are no regulations that could prevent a few companies from monopolizing the accessible supply. Private firms also complicate questions about technology sharing. Zhang et al. suggest one way to prevent a cartel from forming is to open pathways for the exchange of aerospace technology that could facilitate asteroid mining so that more countries could benefit from this industry (9). But whereas technology sharing agreements between governments, while far from cosmopolitan, face fewer legal barriers than attempting to compel a firm to give up technology that could give them such a vast competitive advantage especially when, as I mentioned, there is a large incentive to control the supply of off-world resources.

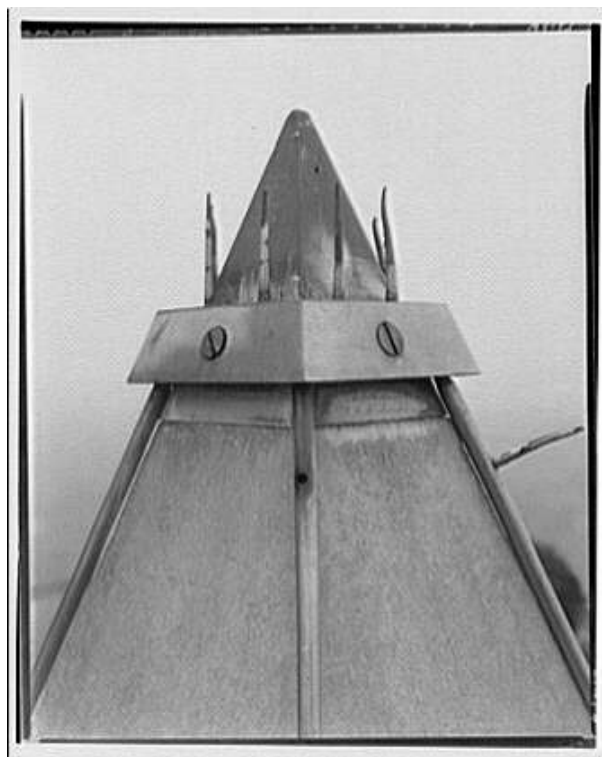
Unfortunately, the solutions for preventing an asteroid-mining cartel are sparse. Much of this is because, as is the case in many innovative

fields, private enterprises outpace regulators. The Declaration on Space Benefits that asserted space activities should be “carried out for the benefit and in the interest of all countries” was made in 1966, three years before the moon landing (10). The laws that govern commercial space activity, like the 2015 SPACE Act, remain vague in their rules, providing only loose guidelines. This is because we still lack a clear understanding of what private space-based EIs will look like. It is still very possible that the exorbitant variable costs of mining asteroids and transporting those resources back to Earth will remain too high for an actual natural monopoly to form (11). Without any idea of what the market might look like, it is too difficult to establish effective regulation. This makes it particularly difficult to form effective regulations that turn prospective firms away from the industry, especially since early firms will be assuming a monumental amount of risk in taking the first steps toward an entirely new, and potentially hazardous, industry.

### **The Consequences of Accessing Off-World Resources**

To some, an asteroid mining cartel that limits supply to control price, could be the best-case outcome. Headlines point to the \$10,000 quadrillion valuation and then the estimated \$100 billion value of the entire global economy and claim that harvesting all its resources would collapse the entire international market. It is very unlikely that, without a cartel, Psyche’s deposits would ever sell at \$10,000 quadrillion on the market. Accessing a reserve of that size will probably crash the entire nickel market, sending its value plunging. Yet, these analysts seem to conflate a disaster in the nickel market with a disaster for the world economy as, despite also boasting an impressive gold deposit, Psyche contains not a single speck of cold hard cash. Therefore, mining Psyche also means hemorrhaging its earth-shattering value. But this does not mean harvesting it would be a fool’s errand. A mission such as that could also be incredibly beneficial to all of humanity. To understand how, let us look at a rare earth metal that we all probably have a roll of in our kitchen, aluminum.

Despite its abundance, aluminum was particularly tricky to refine making it incredibly rare. Napoleon III, who was, among other things, a big fan of aluminum, impressed his guests with his aluminum cutlery, though the less honored ones only got gold. The metal was so valuable that the aluminum pyramid that capped the Washington monument in 1884 cost about \$7000 in today's money, despite only weighing 2.85 kilograms (12). However, only a few years later, Charles Hall and Paul Heroult both separately developed a process using electrochemistry to create purified aluminum. As a result, the price of the metal dropped from about \$12 a pound in 1880 to 78 cents a pound in 1893 (13). However, while aluminum lost its rare earth metal status as a result of the Hall-Heroult method, it found a higher calling than making up a failed monarch's cutlery set. On one hand, aluminum could now be used in common items like storage and cookware. Though more excitingly, the prevalence of aluminum as a cheap material paved the way for its use in new flight technology. It was aluminum that encased Kitty Hawk's crankshaft and it was Duralumin, an aluminum alloy invented in 1909, that proved to be both durable and lightweight enough to be used in metal aircraft (14). It would have been inconceivable to think that a rare metal could have been used to such an industrial extent.



*Aluminum was also selected so the cap could act as a lightning rod. Though after a thunderstorm in 1885 caused small cracks in the structure, copper rods were added (15).*

Mining asteroids for goods could also have profound impacts on terrestrial economic development that might not even be realized until the resources on those rocks can be accessed. Scientist Philip Metzger at the University of Central Florida suggests that the abundance of minerals in asteroid mining could introduce the global market to significantly improved energy collection methods, such as opening the possibility for technologies that could directly harness solar energy via the microwave (16) But why stop there? In 1903, Russian rocketeer Konstantin Eduardovich Tsiolkovsky set out 14 points that mankind should follow to expand beyond Earth. Number 12 is of interest here: “exploitation of asteroid resources to achieve autonomy from Earth” (17). While we have not followed his guide to the letter, Tsiolkovsky was on to something when he said that the purpose of mining asteroids is to achieve autonomy from the terrestrial economy altogether. This is the type of thinking that companies interested in space-borne resources could benefit from. So many analysts approach the industry as a means to access more supplies for the terrestrial economy, but successfully mining asteroids could revolutionize the market by establishing a foothold for the first large-scale economic activity in space.

## **Conclusion**

It is a fact that processes that can access contingency resources threaten to devalue that resource on the market. It would certainly be a disaster for the copper industry if a private enterprise were able to flood the market with celestial sources. But the potential to revolutionize the economy is also great. Making had long dreamed of flight, but it was accessible aluminium that made it the reality it is today. Being able to access material from asteroids could create an entirely new market for those goods, a market that reaches far beyond the Earth’s gravitational pull.



Though there remains a significant risk that a few powerful firms could monopolize this new form of resource extraction, establishing an entirely private market far from the reaches of government regulation. This is the stuff of dystopia, and we currently lack any mechanism to prevent this from happening. On the other hand, overactive regulators could significantly hurt private firms' ability to take steps towards the cosmos, leaving us vulnerable to the dire circumstances that necessitate expansion in the first place. Any tangible suggestions are far beyond the scope of this paper, leaving me to conclude only with a warning that our actions in near future could determine the path of humanity for centuries to come.

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